

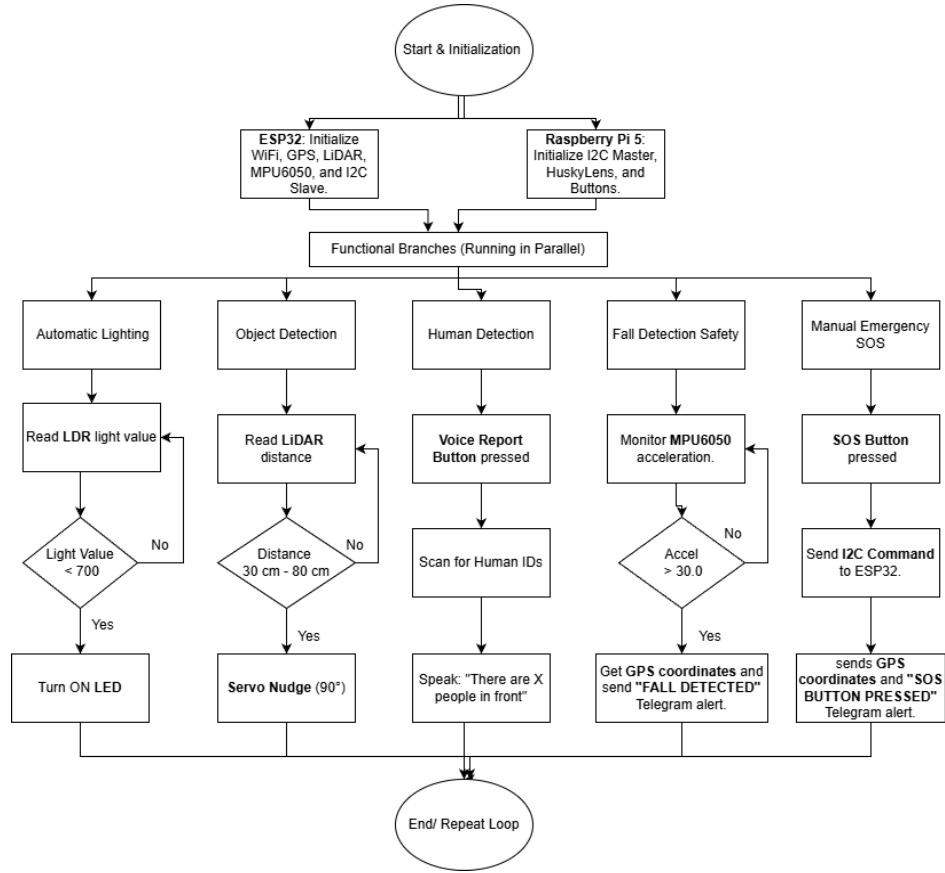


Fig. 3 Flowchart explaining the work principle of IntelliCane

2.2 Schematic Diagram and Prototype Setup

Fig. 4 is the prototype design of the IntelliCane which highlights the mechanical setup and component placement. It illustrates the mounting of the components, alongside the actual internal arrangement of the Raspberry Pi 5 and ESP32 within the custom enclosure. In the meantime, Fig. 5 exhibits the complete circuit diagram of the system, detailing the wiring connections between the power management unit, the dual-processor architecture, and the peripheral sensors for System 1 (Automated Lighting), System 2 (Emergency Alert), System 3 (Object Detection), and System 4 (Human Detection).

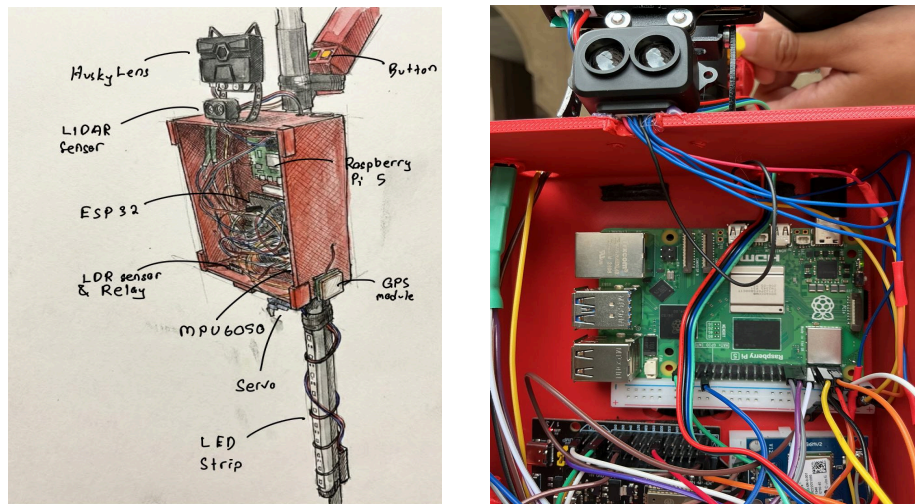


Fig. 4 Prototype Schematics and the actual setup

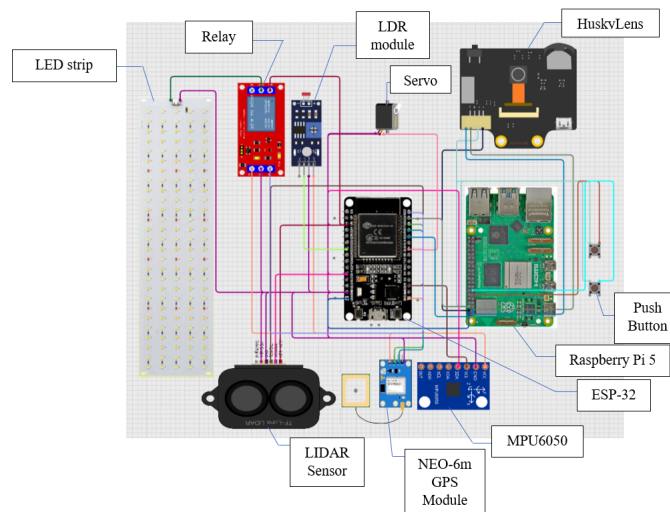


Fig. 5 Circuit Schematics on Cirkuit Designer